

Zedong Peng

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Education

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|-------------|---|--------------------------------|
| M.S. | Shanghai Jiao Tong University , Computer Science
Advised by Prof. Jieru Zhao. | Shanghai, China
2024 – 2027 |
| B.S. | Shanghai Jiao Tong University , Electronic Engineering
Outstanding Graduate. | Shanghai, China
2020 – 2024 |

Experience

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| Research Intern , <i>National Center of Technology Innovation for EDA</i>
Advised by Prof. Qiang Xu and Prof. Jieru Zhao. | Nanjing, China
July 2024 – present |
| <ul style="list-style-type: none"> LLM post training, data construction on domain specific language. | |
| Machine Learning Intern , <i>Intel Asia-Pacific Research & Development Ltd.</i>
Advised by Yuan Zhou | Shanghai, China
Nov 2023 – July 2024 |
| <ul style="list-style-type: none"> SQL Native Engine (Gluten): optimizing CI pipeline. LLM framework optimizations: improve Intel devices support for vLLM. Building LLM applications: Text to SQL. | |
| Winter School , <i>Data Science Institute, Imperial College London</i> | London, UK
Jan 2024 – Feb 2024 |
| <ul style="list-style-type: none"> Applied vector representations to analyze COVID-19 language modal features. | |
| Research Intern , <i>Shanghai Jiao Tong University</i>
Advised by Prof. Liyao Xiang. | Shanghai, China
July 2022 – July 2024 |
| <ul style="list-style-type: none"> Advanced security-focused machine learning research. | |

Honors and Awards

- Outstanding Graduate of Shanghai Jiao Tong University, 2024.
- 2nd Prize, 13th National University Mathematics Competition, 2021.
- 1st Prize, SJTU "Yunhan Cup" RobotMaster Intramural Competition, 2021.
- Liu Yongling Foundation Scholarship, 2021.
- Advanced Individual in Social Practice of Shanghai Jiao Tong University, 2021.

Publications

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| HIPPO: A Hierarchy-Preserving and Noise-Tolerant Pre-HLS Power Modeling Framework for FPGA
Zefan Lin, Zedong Peng , Mingzhe Gao, Jieru Zhao, Zhe Lin
<i>Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD)</i> | 2025 |
| ForgeHLS: A Large-Scale, Open-Source Dataset for High-Level Synthesis
Zedong Peng , Zeju Li, Mingzhe Gao, Qiang Xu, Chen Zhang, Jieru Zhao
arXiv preprint arXiv:2507.03255 | 2025 |
| DeepCircuitX: A Comprehensive Repository-Level Dataset for RTL Code Understanding, Generation, and PPA Analysis
Zeju Li, Changran Xu, Zhengyuan Shi, Zedong Peng , Yi Liu, Yunhao Zhou, Lingfeng Zhou, Chengyu Ma, Jianyuan Zhong, Xi Wang, Jieru Zhao, Zhufei Chu, Xiaoyan Yang, Qiang Xu
<i>2025 IEEE International Conference on LLM-Aided Design (ICLAD)</i> | 2025 |
| A Verifiable and Privacy-Preserving Federated Learning Training Framework
Haohua Duan, Zedong Peng , Liyao Xiang, Yuncong Hu, Bo Li
<i>2024 IEEE Transactions on Dependable and Secure Computing (TDSC)</i> | 2024 |

Projects

Robotic Dog Development

Advised by Prof. Weiming Zhao and Xiaomi Technology.

- Implemented algorithms that improved locomotion control and user interaction.

Shanghai, China
Nov 2023 – Nov 2024

VR Game "Strange Zoo"

Original VR adventure puzzle game developed for CS085 VR & AR design practice.

- Built interactive gameplay mechanics in Unity3D with C# and Microsoft VR tooling.

Shanghai, China
Sept 2021 – Jan 2022

Robot Cart Design and Control

SJTU "Yunhan Cup" RobotMaster Intramural Competition champion project.

- Designed and fabricated a competition robot with Arduino-based control and C programming for precise actuation.

Shanghai, China
Sept 2021 – Jan 2022

Skills

Programming: C++, Python, SQL, Verilog, Shell, Docker, MATLAB

Languages: Chinese (Native), English (CET-6 481), German (A1)